PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Soft-shell Headgear, Concussion and Injury Prevention in Youth Team Collision Sports: A Systematic Review
AUTHORS	Makovec Knight, Jennifer; Nguyen, Jack; Mitra, Biswadev; Willmott, Catherine

VERSION 1 – REVIEW

REVIEWER	Valentina Gallo University of Groningen, Campus Fryslan The Netherlands
REVIEW RETURNED	11-Oct-2020

GENERAL COMMENTS	This is an interesting systematic review assessing the evidence of soft headgear in relation sport-related concussion and other traumas, in youths. It is interesting and overall sound, but lacks some methodological robustness in particulat in relation to report results and incorporate the critical appraisal of the single paper evidence into the overall data interpretation. More detailed comments as follows: Abstract - What is a “systematic review strategy”? - What is the comparison group? This needs to be specified since the abstract Strengths and weaknesses: - This sentence “A literature search revealed only five studies that specifically pertained to youth, rendering the findings less rigorous due to the potential for the adult participants to have diluted the differences in outcomes” does not make sense Introduction - Why reporting epidemiological data referring to the US only? Is there not any more global data? The review does not specifically focuses on US - What’s “game development” in relation to concussion prevention? - The authors refer to only two previous literature reviews with contrasting findings. However this seems a bit reductive and there are other reviews that (although with slightly different approaches) might be relevant and important quoting for providing a fair picture of the background information Methods - Despite the author state that they have followed the PRISMA guidelines there is not reference to a protocol (registered or unregistered, published or unpublished) and inclusion/exclusion

	criteria are not specified comprehensively. I strongly recommend using the PICO classification of criteria as recommended by PRISMA - In particular, where only randomised controlled trials included or not? If not, which study designs were considered? On what population? Vs which comparison groups? Results - In Figure 1 usually the results from each search engine used are reported separately and then the total minus the duplicated are removed, calculated (please refer to PRISMA) - One study was excluded on the basis that it was published before 1980, however there are no time restriction among the inclusion/exclusion criteria. Was this an a priori decision or post-hoc? - In Table 1, row 1: I assume headgear use is the main exposure variable rather than the covariate. If the study was a RCT you would not expect to have covariates - Table 1: when reporting RCT would be useful to know something about randomisation. Given that these could not be double-blinded RCT it is important to assess carefully on which basis boys and girls were assigned to wear or not to wear a headgear, and what was the compliance of randomisation - Table 1: similarly, in observation studies it would be interesting to have data about context in which some youth decided to war a headgear and other did not. Additionally it is very relevant for cohort studies to know if athletes consistently wore headgears. For example, were athletes either waring or not wearing them, or could switch from one exposure group to another each match? There is some potential for bias in this second case as the athlete could decide to wear a headgear depending on his she/he felt on the day which in turn could be associated to his/her propension to risk - Table 1: in all cases, the unit of analysis and the comparison group need to be reported. Can you say that in all 8 papers you compared the incidence of SRC among those wearing a headgear with the incidence of SRC among those not wearing it? - Table 2: this table needs some more work. The classification in columns is excellent, but not always the right data are reported in the cells. In the columns “No-HG exposures”, “HG exposures”, “No-HG concussion”, “HG concussion”, “No-HG superficial head injury”, “HG superficial head injury”, “No-HG all body regions combined” and “HG all body regions combined” I would expect only absolute number where available as for the first row. You might want to add columns (or an additional row per study) for the effect size (for example “HG associated to concussion” or “HG associated to superficial head injury”). Moreover, relative measures need to specify if they are risk rate ratios or relative risks and all relative effect size should be accompanied by the 95% C.I. (as in row 3) otherwise they are uninterpretable. Once this is done, you can get rid of the last column, as the judgment of significant/non-significant should be left to the reader interpreting the reported data - Table 2: I assume that when reporting the study by Deleney you report the same RR flipping the exposure groups on both sides of the fraction. I would also discourage you to do so because inverse ratios do not add anything and might be confusing. - Table 2: it is very interesting seeing how the exposure is quantified (per person, per week, per hour) I believe this deserves a lot more attention and should be added in Table 1. Also a careful reflection on how the different quantification might introduce bias or error would be welcome

	- Even if the studies report incidences in different way, I would recommend try and make the reporting in your result section more consistent. For example you can work out crude incidence ratios where incidence rates are reported (as for reference 23). Please always provide 95% C.I. instead of p-value next every estimate of effect - The authors of these review have the authority of re-assess the evidence, there is no need to say separately what conclusions came the authors of the original paper and then reporting a contrasting figure as for HR and superficial head injury - In general whenever reporting a finding in the results section, including the non-association finding, I would expect to find: an effect estimate with an explicit reference to what is compared to what and relative 95% C.I. as minimum. - At some point in the result section you mention poor compliance as a possible explanatory reason for a result. Where is the critical appraisal of all the studies? What is their potential for bias and confounding, and how might have this biased the results and affected your overall interpretation of data? This set of results should be presented earlier in the text and the author should navigate the reader in reading results in an overall cautious way Discussion - The discussion around heterogeneity of definition of concussion and other outcome measures is very good - A contextualisation of results is missing. How do these results relate to the adults wearing HG for example? Has any research being carried out on subjective perception of HG utility and acceptance among players? Conflict of interest - The author should openly declare any potential conflict of interests

REVIEWER	Arthur Maerlender University of Nebraska Lincoln, USA
REVIEW RETURNED	16-Oct-2020

GENERAL COMMENTS	The focus of this systematic review was on in-vivo evidence of the effect of soft-shell headgear in youth contact sports. Sadly, little in-vivo empirical research exists so this important topic is poorly studied. This review is very well done and follows multiple guidelines. The authors have attended to the necessary elements and provide a compelling narrative in support of their conclusions. We have no concerns with the methodology and in fact, applaud the authors for their rigorous approach. The authors' points about risk compensation behaviour are very important. Yet we would argue that poor mechanics are at least as problematic as wearing headgear. Recent unpublished bench data does show a mitigation of rotational and linear forces, especially at ball velocities under 25 m/s.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Reviewer Name: Valentina Gallo
Institution and Country: University of Groningen, Campus Fryslan, The Netherlands
Please state any competing interests or state 'None declared': None declared

Comments to the Author
This is an interesting systematic review assessing the evidence of soft headgear in relation sport-related concussion and other traumas, in youths. It is interesting and overall sound, but lacks some methodological robustness in particular in relation to report results and incorporate the critical appraisal of the single paper evidence into the overall data interpretation
Author Response: Thank you for your valued input which we have addressed below.

Abstract
Associate Editor comments
- Please state the search dates in the Abstract.
- Please list the search engines in the methods section of the abstract
Reviewer 1 comments
- What is a "systematic review strategy"?
- What is the comparison group? This needs to be specified since the abstract

Response: We have included further detail in the Abstract that includes search dates, 95% CI, databases used, comparison group, and amended "systematic review strategy" to "a systematic search". Sample text [page 2]… "A systematic search in Ovid MEDLINE, Cochrane Library, Scopus, PsycINFO, and SPORTDiscus was conducted in April 2020. Inclusion criteria were youth <18, English language, in-vivo studies published after 1980 that evaluated SRC and other injury incidence in HG users compared to non-users."
"The majority of studies (n=5) reported no difference in the rate of SRC among HG users versus non-users. Contrasting results were seen in two studies, with one in rugby indicating significantly lower risk of SRC for non-HG users (RR: 0.63; 95%CI: 0.41-0.98) compared to HG users, and a cross-sectional survey of soccer players indicating more than double the risk of SRC for non-HG users (RR: 2.65; 95%CI: 1.23-3.12) compared to HG users. Three of the four studies investigating superficial head injury found no significant differences with HG use, though the soccer survey reported reduced risk among HG users (RR= 1.86; 95%CI:0.09- 0.11). Increased incidence of injuries to all body regions for rugby HG users was reported in two studies with adjusted RRs of 1.16 (95%CI: 1.04-1.29) and 1.23 (95%CI: 1.00-1.50)."
- Please ensure that the information provided in your article is consistent with that included in PROSPERO. For example, the search engines. Please update the manuscript and/or PROSPERO accordingly.
Response: PROSPERO has been updated with a PICO statement (Population- In youth team sports, Intervention- Is padded headgear compared with, Comparison- no padded headgear, Outcome- Associated with the frequency of superficial head injury, concussion, and injuries elsewhere on the body?), search engines used (Ovid MEDLINE, Cochrane Library, Scopus, PsycINFO, and SPORTDiscus. and tools used for risk of bias assessment Newcastle-Ottowa Risk scale, the 11-item Physiotherapy Evidence Database (PEDro) scale and the Oxford Centre for Evidence-Based Medicine (OCEBM) guidelines. Authors are awaiting on approval from PROSPERO editorial team for these changes to be publicly visible.
Strengths and Weaknesses:

- Please ensure that you have fully described the methodological limitations of the study in the strengths and limitations section.

- This sentence “A literature search revealed only five studies that specifically pertained to youth, rendering the findings less rigorous due to the potential for the adult participants to have diluted the differences in outcomes” does not make sense

Response: Thank you for raising these points, we have added additional points regarding limitations. Sample text [page 3]...”The literature search revealed few articles. Included studies generally lacked robust evidence and did not assess the intervention of headgear as the primary outcome. As such, the conclusions should be interpreted with caution.” Sample text [page 17]... “Without robust data it was difficult to draw definitive conclusions about the role of HG in sports injury prevention. An important methodological issue was that reporting of results across studies was inconsistent. Reporting of homogenous outcomes and 95% confidence intervals was not possible in all cases as data was not available, although attempts were made to re-analyse some data to provide consistency. “And we amended the sentence with poor grammar. “The review included only five studies that specifically pertained to youth cohorts. Many studies combined adult and youth participants, potentially confounding findings regarding risk taking behaviour with headgear which may differ across age.”

Introduction

- Why reporting epidemiological data referring to the US only? Is there not any more global data? The review does not specifically focuses on US

Response: Yes, this is certainly a valid point. This oversight has been amended to include global data. Sample text [page 4]…”Traumatic brain injury (TBI) is a leading cause of disability and death, contributing to a growing worldwide disease burden. Global estimates indicate that TBI affects 60 million individuals per year.”

- What’s “game development” in relation to concussion prevention?

Response: The authors were referring to the idea that sporting policy has used rule changes and altered training practices to improve concussion prevention. This was unclear and has been amended to include detailed explanation of what this entails. Sample text [page 4]... “Playing technique (e.g. teaching skills that reduce exposure to head impacts), rule changes (e.g. limiting tackling and contact) and the use of protective equipment (e.g. hard shell helmets, padded headgear and mouthguards).”

- The authors refer to only two previous literature reviews with contrasting findings. However, this seems a bit reductive and there are other reviews that (although with slightly different approaches) might be relevant and important quoting for providing a fair picture of the background information

Response: Despite the limited scope of published findings within this field, a broader view has been provided with two additional references added; one systematic review of sports concussion interventions (including headgear) Emery, Black, Kolstad et al. (2017) and another highlighting current opinion on whether headgear is effective as sports concussion prevention Gardner, Quarrie, Iverson GL (2020).

Methods

- Despite the author state that they have followed the PRISMA guidelines there is not reference to a protocol (registered or unregistered, published or unpublished) and inclusion/exclusion criteria are not specified comprehensively. I strongly recommend using the PICO classification of criteria as recommended by PRISMA
- In particular, where only randomised controlled trials included or not? If not, which study designs were considered? On what population? Vs which comparison groups?

Response: A protocol was registered with PROSPERO and reference to this has been added to the methods. PICO classification of criteria has been added with detail included population., comparison groups and study designs considered (RCTs, cross sectional and observational injury surveillance) Sample text [page 6]... “Studies were included when the population of interest were athletes under aged 18 years who participated in a collision team sports study assessing the intervention of HG in comparison to no-HG upon the primary (SRC) and secondary outcomes (head injury superficial to the skull and/or injuries to other body regions). Included studies were limited to those using quantitative methods to report SRC, head injury and other injury rates (e.g., cross sectional survey, prospective cohort injury surveillance, randomized control trials).”

Results

- In Figure 1 usually the results from each search engine used are reported separately and then the total minus the duplicated are removed, calculated (please refer to PRISMA).
Response: As suggested Figure 1 has been amended to include all search engines and results. This was not completed as track changes as it was difficult to configure.
- One study was excluded on the basis that it was published before 1980, however there are no time restriction among the inclusion/exclusion criteria. Was this an a priori decision or post-hoc?
Response: This was an a priori decision that the authors made an error in initially omitting. Exclusion criteria have been amended to include this. Sample text [page 6]... “Studies were excluded if they were published prior to 1980, laboratory based, conducted in adult only cohorts, conducted in individual and/or non-contact sports or only included participants wearing hard-shell helmets.”

- In Table 1, row 1: I assume headgear use is the main exposure variable rather than the covariate. If the study was a RCT you would not expect to have covariates
Response: That is correct. Row one amended to exclude this.

- Table 1: when reporting RCT would be useful to know something about randomisation. Given that these could not be double-blinded RCT it is important to assess carefully on which basis boys and girls were assigned to wear or not to wear a headgear, and what was the compliance of randomization
Response: We agree that this is an important point. We have added a row in table 1 titled “Randomisation of HG use, compliance and wearing rates” with extracted data from each study included (where available). Attempts have also been made to consider this point in the critical appraisal of the methodology of included studies.

- Table 1: similarly, in observation studies it would be interesting to have data about context in which some youth decided to war a headgear and other did not. Additionally, it is very relevant for cohort studies to know if athletes consistently wore headgear. For example, were athletes either wearing or not wearing them, or could switch from one exposure group to another each match? There is some potential for bias in this second case as the athlete could decide to wear a headgear depending on his she/he felt on the day which in turn could be associated to his/her propension to risk
Response: We agree that this is a flaw in methodology in many of the included studies where information on HG wearing rates was seldom reported. These data have been extracted where available and detail provided on the methods used to quantify headgear use (e.g., one study used a Likert measure, others used a single time point survey and others were assigned to wear HG).

- Table 1: in all cases, the unit of analysis and the comparison group need to be reported. Can you say that in all 8 papers you compared the incidence of SRC among those wearing a headgear with the incidence of SRC among those not wearing it?

Response: Our best efforts have been made to include data in both HG and No-HG groups where available. Amendments have been made to include units of analysis so that incidence of SCR in each group (HG and No-HG) is now present, with the exception of the one study that did not analyse this outcome (Chalmers et al., 2011). That study reported "all injury rates" only.

- Table 2: this table needs some more work. The classification in columns is excellent, but not always the right data are reported in the cells. In the columns "No-HG exposures", "HG exposures", "No-HG concussion", "HG concussion", "No-HG superficial head injury", "HG superficial head injury", "No-HG all body regions combined" and "HG all body regions combined" I would expect only absolute number where available as for the first row. You might want to add columns (or an additional row per study) for the effect size (for example "HG associated to concussion" or "HG associated to superficial head injury"). Moreover, relative measures need to specify if they are risk rate ratios or relative risks and all relative effect size should be accompanied by the 95% C.I. (as in row 3) otherwise they are uninterpretable. Once this is done, you can get rid of the last column, as the judgment of significant/non-significant should be left to the reader interpreting the reported data

Response: Thank you for the suggested corrections. Absolute numbers have been extracted and provided in all cases where possible. Unfortunately, some studies preferred to report incidence rates or rate ratios and no effect sizes were reported. We have amended the table to outline which studies used incidence rate ratios or relative risks. In addition, we have calculated crude relative risks with 95% CI for all studies that reported the data needed to complete the analysis. The studies that reported adjusted relative risk were not re-calculated as the findings are now much easier to interpret across studies. As suggested, the last column outlining the "findings" has been deleted. P-values were not extracted as they were not available in many studies, and the reader should be able to interpret the reported data now with inclusion of all 95% CIs. The one exception was that we could not calculate relative risk for Archbold et al.,(2017) for the superficial head injury outcome. These authors provide a "log rank statistic" from a cox proportional hazard model. We did not have the data to calculate a Relative Risk here, and as it was not a primary outcome of the review, the statistic has been left as the original authors reported, with that findings explained in text.

- Table 2: I assume that when reporting the study by Delaney you report the same RR flipping the exposure groups on both sides of the fraction. I would also discourage you to do so because inverse ratios do not add anything and might be confusing.

Response: This has been amended to only include the Relative Risks as reported by the original authors with no flipping of exposure groups. In addition, we used the available data to calculate 95% CI's which were not reported by Delaney et al.

Table 2: it is very interesting seeing how the exposure is quantified (per person, per week, per hour) I believe this deserves a lot more attention and should be added in Table 1. Also, a careful reflection on how the different quantification might introduce bias or error would be welcome

Response: These details have been added to Table 1 and reflected upon in the critical appraisal of studies in the results section. Sample text [page 10]... "The most common convention was for authors to report injuries per player 1000 player games. The two studies who did not conform to this, were subject to retrospective self-report bias. For instance, Marshall et al., (2005) obtained injury data from players at weekly intervals and therefore reported by 'player week', and Delaney et al., (2007) reported injuries "per player", likely because players reported SRC at a single survey time point. These studies likely reported injuries in this way because they did not capture players exposure to injury, and therefore could not report per hour."

- Even if the studies report incidences in different way, I would recommend try and make the reporting in your result section more consistent. For example, you can work out crude incidence ratios where incidence rates are reported (as for reference 23). Please always provide 95% C.I. instead of p-value next every estimate of effect.

Response: Crude relative risks have been calculated with an asterisk to indicate where this was done. Details on some limitations to this process are outlined in above response.

- The authors of these review have the authority of re-assess the evidence, there is no need to say separately what conclusions came the authors of the original paper and then reporting a contrasting figure as for HR and superficial head injury.
Response: Thank you for highlighting this, the sentence containing the contrasting finding has been amended to reflect this, and it has been considered more generally.

- In general, whenever reporting a finding in the results section, including the non-association finding, I would expect to find: an effect estimates with an explicit reference to what is compared to what and relative 95% C.I. as minimum.
Response: We have edited the text to include these statistics with reference to the comparison groups where appropriate. All other data are located in Table 2. Sample text [page 11]... "Contrasting findings were seen in two other studies; a prospective cohort study in rugby showed that non-HG users were at significantly lower risk of SRC (RR: 0.63; 95%CI: 0.41-0.98) than HG users, and a cross-sectional survey of soccer players outlined higher risk of SRC for non-HG users (RR: 2.65; 95%CI: 1.23-3.12) compared to HG users."

- At some point in the result section you mention poor compliance as a possible explanatory reason for a result. Where is the critical appraisal of all the studies? What is their potential for bias and confounding, and how might have this biased the results and affected your overall interpretation of data? This set of results should be presented earlier in the text and the author should navigate the reader in reading results in an overall cautious way.
Response: We have moved the critical appraisal of studies to immediately after Table 1, so the reader has context and a basis for evaluating the studies. We have added a paragraph outlining issues with poor compliance , poor recording of actual HG use and the bias introduced by quantifications of exposure. We have directed the reader to interpret results with caution. Sample text [pages 10 & 11]..."Overall, methodology across studies tended to lack scientific rigour in one or more aspects..."

Discussion
- Please include subheadings in the Discussion section that would help guide the reader.
Response: Thank you for this suggestion. These have been added where relevant. Sample text [page 13]... "The Association of Headgear with Sports Related Concussion, Superficial Head Injury and other Injuries. "

- The discussion around heterogeneity of definition of concussion and other outcome measures is very good
Response: Thank you. We aimed to provide a thorough analysis of these issues.
- A contextualisation of results is missing. How do these results relate to the adults wearing HG for example? Has any research being carried out on subjective perception of HG utility and acceptance among players?
Response: Research has been carried out on the perceptions of wearing HG. We have added an additional paragraph outlining some of these findings with reference to how the findings from the perception research, and findings from the review may interact to impact the generalizability of results to adult HG users. Sample text [page 14].."Given that perceptions about HG and associated behavioural changes may differ across the lifespan, it is unclear whether injury risk associated with HG use differs between adult and youth populations."

Conflict of interest
- The author should openly declare any potential conflict of interests

Response: Have amended accordingly. Sample text [page 18]... “Conflicts of Interest: None declared. “

Reviewer: 2
Comments to the Author
The focus of this systematic review was on in-vivo evidence of the effect of soft-shell headgear in youth contact sports. Sadly, little in-vivo empirical research exists so this important topic is poorly studied.

This review is very well done and follows multiple guidelines. The authors have attended to the necessary elements and provide a compelling narrative in support of their conclusions. We have no concerns with the methodology and in fact, applaud the authors for their rigorous approach.

The authors’ points about risk compensation behaviour are very important. Yet we would argue that poor mechanics are at least as problematic as wearing headgear. Recent unpublished bench data does show a mitigation of rotational and linear forces, especially at ball velocities under 25 m/s.
Response: Thank you for this thoughtful feedback, we look forward to seeing more published data on this important topic.